

MSC 487, Spring 2025-2026

Introduction to Electroacoustic Music

Professor: Tolga Yayalar (tolga.yayalar@bilkent.edu.tr)

Class Hours: Monday 15:30-17:20, Thursday 10:30-12:20 (room: 341)

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DESCRIPTION

Basic acousmatic music composition techniques include recording, sound editing, creative sound design, temporal and spectral sound processing, DAW, sampling, automation, modulation techniques, and spatialization, including high-order ambisonics. Projects realized using recording gear and computers; study of relevant aspects of acoustic and electroacoustic music theory; repertoire since 1948 of musique concrète, acousmatic, and tape music; Hands-on work will culminate in a final performance of individual projects.

This course focuses on electroacoustic music as a concert music genre. It follows the tradition that started with composers like John Cage, Edgar Varèse, Pierre Schaeffer, Karlheinz Stockhausen, Luigi Nono, Luciano Berio, Iannis Xenakis, Bernard Parmegiani, Luc Ferrari, Trevor Wishart, Dennis Smalley, Jonty Harrison etc... The primary outcome of the course is to create a multi-channel piece of music along the same aesthetic and technical direction. We will specifically work with high-order ambisonics to realize final projects. You will also have a chance to mix your pieces in binaural audio.

There's no prerequisite in this course; however, it is highly recommended that students have:

- Some knowledge about music theory: at least basic knowledge of notes, scales, chords, rhythms
- Some experience in music-making: piano, violin, etc.. lessons, rock band member, home studio compositions etc...
- More importantly, open-mindedness about art and music. and a certain curiosity about new things and experiences.

What this course **is not** about:

- Making beats, grooves, and loops
- Learning to mix your songs
- Learning about music production and engineering
- Learning to make music with software instruments
- Learning music theory to be able to make better songs
- Producing songs
- Songwriting

What this course is about:

This course focuses on electroacoustic music as a concert art form, with an emphasis on acousmatic composition, critical listening, and sound-based musical thinking. Students will create fixed-media and multi-channel electroacoustic works using recorded sound, signal processing, and spatialization, grounded in the post-1948 electroacoustic tradition.

Audio technology in this course is treated not as a production or songwriting tool, but as a creative medium. Techniques such as editing, sampling, filtering, compression, modulation, automation, and spatialization are explored as compositional processes, often used in unconventional ways to shape form, texture, gesture, and perception rather than to polish songs.

The course includes:

- Composing original works in the tradition of musique concrète, tape music, and acousmatic music
- Developing critical listening skills, with focus on timbre, space, gesture, and texture
- Studying key electroacoustic repertoire and compositional approaches
- Learning fundamentals of acoustic and electroacoustic music theory relevant to sound-based composition
- Working with time, form, and space, including stereo, multi-channel formats, and high-order ambisonics
- Using Reaper as a compositional environment for electroacoustic music

The course culminates in the creation and presentation of a multi-channel electroacoustic composition, with opportunities for binaural rendering.

We will use Reaper to complete all projects and assignments in this course. The focus will be exclusively on producing fixed-media concert pieces. Software instruments, third-party samples, synthesizers, loops, and beats are strictly prohibited. Students are required to generate all sonic material themselves, with absolutely no exceptions. The course is dedicated to electroacoustic music within the context of Western Art Music, and as such, popular genres such as electronica, drum and bass, ambient, dub, synth-pop, techno, trance, EDM, house, and similar styles will not be covered in the class or accepted as project submissions.

This course will not cover synthesis; we will not use synthesizers to create music. This decision is not based on aesthetic preferences but on technical considerations. Synthesis is a broad and complex subject requiring at least a full semester to understand the principles and techniques involved thoroughly. Moreover, a dedicated course explores this topic in depth: COMD 527 Creative Sound Design and Synthesis.

COURSE STRUCTURE.

This is a high-intensity course structured around three interlocked modules.

- 1.** Basic Theory: Although this is primarily a practice-oriented course, a deeper understanding of the concepts is indispensable when working with the electroacoustic medium.
- 2.** Music Creation: Our main purpose of the class is to produce musical works. Students are expected to compose a considerable amount of music almost every week. The first two-thirds of the semester will be spent on the nuts and bolts of basic audio techniques. We will follow a strict curriculum. We will focus on individual projects and more compositional methods for the remaining third of the semester
- 3.** Aesthetic/Literature/Culture: Besides compositional activity, there will also be an emphasis on the aesthetics and culture of electroacoustic music. We will have a weekly listening list and spend about half an hour each week discussing various aspects of the genre. The midterm exam is a listening exam, where you need to identify repertoire pieces that we have covered in class.

Throughout the first half of the semester, the week will be divided into two. On Mondays, we will talk about the listening list, have the student presentation, and cover some theoretical and aesthetic topics. Thursdays will be reserved for theory and also some practical work.

REQUIRED TEXTBOOK

Roads, Curtis: Composing Electronic Music: A New Aesthetic

You must get a copy of the textbook. Most of the readings will be from this book.

SOFTWARE

Since we are writing music for loudspeakers, it is always best to work in the studio. Nevertheless, making electronic music requires much time, so you may also want to use your own computers. We will exclusively use Reaper for everything. It will be our primary tool. This is powerful software that is practically free. I encourage you to download it on your computers as soon as possible and start learning it.

The second software that we will use extensively is the FabFilter plugin suite. This is installed on the studio computer. This is paid software. You don't have to purchase it.

Finally, we will use the IEM suite, also free spatialization software. You can download it through the link below.

You may be used to other DAWs, but in this class, you must use Reaper for all your projects.

Here are the links

Reaper (<https://www.reaper.fm/>)

IEM Plug-in Suite (<https://plugins.iem.at/>)

You must have Reaper and IEM installed on your computer. You must work with Reaper to realize all your projects, including your final piece for the multi-channel system.

The software has a long trial period; it is only 60€ if you want to buy the license. For your projects, I suggest you use the studio. There is some special software that it would be difficult to find elsewhere. Since your final projects need to be for an immersive setup, you must regularly work in the studio for the final project.

For detailed information, please visit:

<https://www.reaper.fm/>

SOFTWARE TO BE INTRODUCED IN THIS COURSE AND AVAILABLE IN THE STUDIO

Reaper

FabFilters

IEM

Kontakt

iZotope RX

GRM Tools

IRCAM Tools

Soundtoys

Waves Plugins

Sound Particles

Ircam Spat Revolution

OTHER TEXTS.

- Cipriani & Giri, Electronic Music and Sound Design: Theory and Practice with Max 7 Volume 1 and 2 (There's great theoretical information in these books. These will be our textbooks at MSC 587)
- Roads, Curtis. The computer music tutorial. MIT press, 1996.
- Loy, Gareth. Musimathics: the mathematical foundations of music. Vol. 1. MIT press, 2011
- Holmes, Thomas B. Electronic and experimental music: pioneers in technology and composition. Psychology Press, 2002.
- Manning, Peter. Electronic and Computer Music, Oxford University Press, 2013
- Wishart, Trevor, On sonic art. Hardwood Academic Publishers, 1996.
- Schafer, R. Murray. The soundscape: Our sonic environment and the tuning of the world. Simon and Schuster, 1993.

SOME ONLINE RESOURCES

Computer Music Composition and Production links (UC Irvine)

<http://music.arts.uci.edu/dobrian/w18/music215A/links.htm>

Music and Computers, Phil Burk

<http://sites.music.columbia.edu/cmc/MusicAndComputers/>

ReaperTutorials

<http://reaper.fm/videos.php>

Projects

There will be five small-scale projects throughout the semester. You will be given some compositional constraints. We will have a class meeting dedicated to these projects. You will have a chance to present your piece and a feedback session afterward.

The final project is going to be a multi-channel concert tape piece written for our 4.8.4.1 system. We will have a 2-day event in June; these pieces will be performed then. All projects must be individual; collaborative projects are not allowed.

The projects must be turned in by the deadline. No late submissions and no exceptions.

Presentation

Every student will be assigned a classic example of electronic music. They will present them in the class. You need to prepare a proper presentation, including handouts with a complete bibliography. The presentations should be 10-minutes

Here's the list of presentations:

January 29th - Reich – Come out (Example)

February 12th - Schaeffer – 5 études de Bruits

February 19th - Edgar Varese – Poème Electronique

March 3rd - Xenakis – Diamorphoses

March 26th - Stockhausen – Gesang der Jünglinge

March 30th - Luc Ferrari – Presque Rien No. 1

April 9th - Jean-Claude Risset – Sud

Listening Journal

We will keep a forum for our listening assignments. You must listen in-depth to all the tracks and fill out your listening journal. You are also encouraged to read others' entries and make

comments. This is our forum where we can exchange ideas about the repertoire. I recommend that you listen to the pieces either with high-quality speakers or with headphones. You need to hear the full spectrum to appreciate electroacoustic music. It will be impossible to hear these pieces fully with your computer speakers.

In the listening journal there will be some questions? Use them as a guide for your entry. Do not use the entries as mere replies to these questions, but write your feedback after you thought about these questions.

Listening journals must be done before the class hour on Monday. If you don't fill it for Monday's class, you will have another 3 days until the end of Thursday. But you will only get half point.

For the listening journal, there are no grades, it is yes or no. But nevertheless, if you don't take it seriously and write irrelevant entries, I may give half point or sometimes no point at all.

Midterm Listening Exam:

The midterm exam is a listening-based identification test covering key works from the electroacoustic repertoire studied in class. Students will hear short excerpts (1-2 minutes) from selected pieces and must identify the composer and title of each work from memory. The exam emphasizes critical listening, stylistic recognition, and familiarity with landmark electroacoustic compositions rather than multiple-choice recall

Attendance/Class participation.

If you miss meetings, your attendance grade of 10 points will drop gradually, according to the table below.

1 absence	14 points
2 absences	14 points
3 absences	14 points
4 absences	13 points
5 absences	11 points
6 absences	9 points
7 absences	6 points
8 absences	3 points
9 absences	1 points
10 absences	0 points
11 absences	FZ

Please note: Unexcused absences are disruptive and disrespectful. If you cannot attend the class, please email me in advance.

Grading

Listening Journal (8): %16 (out of 9, there's one bonus)

Small Projects (5): %25 (%5 each)

Presentation: %5

Midterm Listening Exam: %10

Attendance: %14

Final Project %30

FZ Policy

If you do not turn in 2 projects on time for the collective listening sessions, you will receive an FZ. Additionally, if you miss 10 meetings you will also receive an automatic FZ.

		SUBJECT	READING	LISTENING	ASSIGNMENT
1	22/1	INTRODUCTION TO THE COURSE Introduction to the course Introduction to the culture and aesthetics of Electroacoustic Music as a genre		• Natasha Barrett – Microclimates: Wet Face • Hans Tutschku – Klaviersammlung • İlhan Mimaroğlu - Prelude No. 11 • Gilles Gobeil - Ombres, espaces, silences II	
	26/1	MYSTERY PIECES Music vs music! What do we listen to?	•Roads, chapter 2 "Aesthetic Foundations" pp1-13	6 Mystery Pieces!	•Listening Journal No. 1 "Mystery Pieces"
2	29/1	NATURE OF SOUND/DIGITAL AUDIO What is sound, how do we hear? Properties of sound: Amplitude, Frequency, Timbre Psychological Acoustics Digital sampling	Roads, chapter 3 "The Nature of Sound"	• Steve Reich – Come out* • Robert Normandeau –Eclats de Voix • Luc Ferrari – Unheimlich Schön • Berio – Omaggio a Joyce • Adamek –Die Zeit rennt • BONUS: Bob Ostertag - Sooner or Later	•Project No. 1 -Splicing is assigned •Listening Journal No. 2 "Voice and Language"
	2/2	BASIC RECORDING Recording with Reaper Handheld recorders Audio formats Noise Reduction and basic audio restoration with iZotope RX, Low cut filters Introduction to Reaper and basic editing			
3	5/2	STUDIO VITALS REAPER INTRO Sign out of material, sign out of studio time Fundamentals of DAWs			Reaper Tutorials 1-8
	9/2	PROJECTS I - In class listening Introduction to Handheld recorders - Zoom H4n TAPE STYLE EDITING Tape style editing techniques External audio editors - deconstructive editing			Project No. 1 -Splicing is due Project No. 2 -Concrete Etude is assigned

		SUBJECT	READING	LISTENING	ASSIGNMENT
4	12/2	REDUCED LISTENING AND CONCRÈTE SOUND MATERIAL <i>Listening Modes</i> <i>Musique Concrète</i> Acousmatique Situation <i>L'Objet Sonore</i>	Chion, "In search of the Sound Object" pp.18-33 Roads, Ch. 4 "Creating Sound Materials" pages 63-87	• Pierre Schaeffer – Etudes* • Pierre Henry - Variations pour une porte et un soupir • Tod Dockstader - Water Music • Panayiotis Kokoras - Qualia • Jonty Harrison - Rock "n" Roll	•Listening Journal No. 3 "Musique Concrete"
	16/2	TAPE STYLE EDITING Tape style editing techniques External audio editors - deconstructive editing AUTOMATION AND PARALLEL PROCESSING LFO's, basic side-chaining. Working with controllers Basic signal routing Auxiliary channels Serial vs Parallel Processing			
5	19/2	COMPOSING WITH COMPUTERS Beyond lattice based conventions TARTYP and Spectromorphology	Roads ch. 5 "Sound Transformation"	• Horacio Vaggione – Preludes Suspendus II • Alistair MacDonald - Equivalence • Edgar Varese -- Poeme Electronique* • Gilles Gobeil - Les lointains noirs et rouges	•Listening Journal No. 4 " Concrete cont."
	23/2	SOUND DESIGN ESSENTIALS <i>More in-depth work with Reaper</i>			
6	26/2	PROJECTS II - In-class Listening Brief Introduction to Synthesis			Project No. 2 is due Project No. 3 is assigned
	2/3	NOISE AND FILTERS Filters and basic dynamic processing Working with noise Filtering and EQ Dynamic filters Spectral Filtering/Image Filters	Roads, ch. 4 "Creating Sound Materials" page 96-107	• Iannis Xenakis: Diamorphoses* • Jean-François Laporte -Mantra • Gottfried Michael Koenig – Klangfiguren II • James Tenney –Analog #1 (Noise Study)	•Listening Journal No. 5 "Noise"
7	5/3	SAMPLING Basic sampling techniques Working with Kontakt Creating unusual instruments	Roads, ch. 4 "Creating Sound Materials" page 87-95	• Hans Tutschku - Die zerschlagene Stimme • Denis Smalley - Wind Chimes • Yves Daoust - Quatuor • Horacio Vaggione - Schall	•Listening Journal No.6 "Instruments and Sampling"

		SUBJECT	READING	LISTENING	ASSIGNMENT
	9/3	MODULATION and SIDE CHAINING Modulating parameters with oscillators Modulating parameters through side chaining			
8	12/3	DYNAMIC PROCESSORS Compressor, Expander, Gate, Multi-band compressor, Transient enhancers etc...			
	16/3	No class Spring Break			
	19/3	No class Eid			
	23/3	PROJECTS III - In-class Listening			Project No. 3 is due Project No. 4 is assigned
9	26/3	STEREO IMAGE AND VIRTUAL SPACE Panning, Creating movement etc... Delays and Reverbs	Roads Ch. 8 "Articulating Space"	• Stockhausen: Gesang der Jünglinge* • Ake Parmerud - Les Flutes en Feu • Tod Dockstader: Quatermass, Tango • Natasha Barrett –Little Animals	•Listening Journal No.7 "Stereo Imaging"
	30/3	REVERB and DELAYS Different types of Reverbs and delays Creative usage			
10	2/4	SOUNDSCAPE COMPOSITION Schafer, "The Schaeffer vs Schafer Creating a soundscape	Schafer, "The Soundscape" Introduction	• Alan Lamb –Beauty • Luc Ferrari – Presque Rien No. 1* • Jonty Harrison - Sorties • Hildegard Westerkamp - A walk through the city	•Listening Journal No. 8 - Soundscape
	6/4	PROJECTS IV - In class Listening			Project No. 4 is due
11	9/4	WHAT DO WE LISTEN TO? Preparation for the listening exam Emmerson: Language of Electronic Music	Simon Emmerson - The Relation of Language to Materials	• Jean-Claude Risset: Sud* • John Cage - Williams Mix • Denis Smalley - Pools and Currents • Beatriz Ferreyra - L'autr ... Ou Le Chant Des Marecages	•Listening Journal No.9
	13/4	MIDTERM LISTENING EXAM			
12	16/4	MULTI-CHANNEL SOUND Live Diffusion Introduction to multi-channel mixer / Introduction to Ambisonics		• Ake Parmerud - Les Flutes en Feu • Joseph Anderson - Pacific Slope	Project V is assigned

		SUBJECT	READING	LISTENING	ASSIGNMENT
	20/4	MULTICHANNEL WORKSHOP Working with Ambisonics. Creating a HOA version of "Desert Station"			
	23/4	No class: National Sovereignty holiday			
13	27/4	PROJECT V -- In class Listening			Final Project is assigned
	30/4	MICROSOUND Granular Synthesis Sound Particles		• Bernard Parmegiani – De Natura Sonorum II • Maggi Payne – Fizz • Elainie Lillios - Listening Beyond	
14	4/5	SPECTRAL PROCESSING Basic FFT, Introduction to spectral tools, Audiosculpt, Soundhack vs... Sampling Techniques			
	7/5	SONIC NARRATIVE Sonic drawing, spectral shapes, gestures vs..) Creating space in Electronic Music Advanced Editing Tools - Metasynth	Roads ch. 10 "Sonic Narrative"	• Jonty Harrison - Klang • Jonathan Harvey: Mortuos Plango. Vivos, Voco	
	29/5	Final Projects due			

The concert date will be announced later in the semester

MSC 487 Introduction to Electronic Music
PROJECT 1 -SPLICING

Bilkent University — Department of Music — Spring 2026 — Tolga Yayalar

DEADLINE: 9 February during class

Compose a 2-4 minute piece with the following constraints:

1. Record the voice of a person. Ideally, it should be someone else. You can use a text, it could be in any language. Or you can record non-textual voice. You can record different takes and different people, but it must only be a human voice (not AI). The recording must be:
 - 48kHz, 24 bits
 - Mono
2. You should either record this in the studio directly to Reaper, or use a handheld recorder.
3. Prepare your sounds with basic, noise reduction, spectral cleaning, low cut filter etc...
4. Compose your piece by editing the sounds you recorded. In your edits, you are allowed to use only the recording you made. Be creative. You can use all the information in your recording, including the noisy parts, such as breathing noise, vocal fragments, shouting, whistling, etc.
5. No effects or any other treatments. Only cut, copy, paste, and volume adjustments. Except for the treatments that you used to clean up your recording.
6. You can do multitrack layering, loops, surgical editing, basic fade-in and fade outs.
7. When your piece is ready, render it to a mono WAV file and be ready to present in the classroom. Do NOT make it stereo.
8. The piece needs to be ready to present on the project day. Failure to do so will result in a failing grade. If you can't make it to the presentation, then you are graded on 75 points.

GRADING

Mono file: 10 pts

48kHz: 10 pts

24bit: 10 pts

No clipping audio: 20 pts

Rules are followed: 25 pts

Presentation: 25 pts (This is simple if you are in the class and present the composition you get the 25 pts, otherwise no points given)

MSC 487 Introduction to Electroacoustic Music
PROJECT 2 -TAPE PIECE

Bilkent University — Department of Music — Spring 2026 — Tolga Yayalar

DEADLINE: 26 February during class

Compose a 2-4 minute-long piece with the following constraints.

1. With this project, you can with the following material:

- Pick only one of the classical elements: earth, water, air, and fire.
- You may record as many samples as you like within your chosen category. Build a library with your recordings.
- You don't have to limit yourself to one source. You can have as many source sounds as you want within your chosen category. Be creative! For example:

Earth: Weight, friction, impact, granular texture

Digging, shoveling, or raking soil; Stones/rocks/pebbles; Leaves/Branches; Footsteps, Concrete, etc...

Water: Flow, resonance, continuity, articulation, Ice, Snow, Rivers/streams; Raindrops; Pouring, splashing, or dripping water; Bubbling water; Ice cracking; Pipes; Fountain, etc.

Air: Breath, pressure, turbulence, invisibility

Wind; Breathing; Fans/Vents; Blowing; Balloons etc...

Fire: Energy, instability, crackle, transformation

Matchstick; Gas oven; Fireplace; Charcoal burning; Paper/Wood burning; Welding; Torch; Crackle; etc...

2. Transfer sounds to the computer and create a sound library by labeling the sounds accordingly.

- You may want to edit the sounds first, and make sure to clean them.
- Label them accordingly so you know what is what

3. Compose a piece of music using only the sounds you recorded for your library. Make sure the piece has a coherent structure or formal trajectory.

4. You can only use the classic tape music techniques in this project.

- Editing and Splicing
- Volume automation / Panning
- Reverse Playback
- Variable Speed Playback and Pitch Shift (either tape-style or independent)
- Delays
 - Simple Analog Single-tap Delay

- Timeless, ReaDelay
- Multi-tap Analog Delay
 - Waves/SuperTap
- Tape loop
- Tape Echo or Spring/Plate Reverbs
 - EchoBoy, TR5 Tape Echo, Super Plate, TR5 CSR Plate
- Flanging/Phasing
- Basic Reverb - FabFilter R2, ReaVerbate
- EQs
 - FabFilter Pro-Q 4
 - ReaEQ
- Ring Modulation

5. Your piece should adhere to the principles of musique concrète. This means that the sounds in your composition should become disassociated from their real-world sources. Your goal is to create pure sound objects—sounds treated and sculpted for their inherent qualities, not for the objects they originated from. Highly recognizable literal sounds should be transformed enough that their source is no longer immediately identifiable. Think of yourself as a sculptor, shaping and transforming sounds in time to create a work that stands on its own, free from any literal associations. Ensure your piece has a coherent structure or narrative, guiding the listener through a distinct sonic journey. You may use as many tracks as necessary to achieve your desired result.
6. You can use as many tracks as you need.
7. When finished, export your piece to a 48 khz/24bit stereo format wav file.
8. Pieces due on February 26th. Be prepared to play them in the class.

GRADING

Correct Audio format: 10 pts

No digital clipping or unintended distortion: 10 pts

Sound Source rules are followed: 25 pts

Adherence to Musique Concrete Aesthetics: 30 pts

Presentation: 25 pts

MSC 487 Introduction to Electronic Music

PROJECT 3 - SOUND AS INSTRUMENT

DEADLINE: 23 March during class

Bilkent University — Department of Music — Spring 2026 — Tolga Yayalar

This project explores the moment when recorded sound stops being a fixed object and becomes a controllable instrument. The focus is on repetition, variation, gesture, and pattern—not on storytelling or realism. Instead of treating recordings as finished sounds, you are asked to turn them into material that can be played, shaped, and controlled over time. By building a sampled instrument, you will explore how sound can respond to gestures such as dynamics, register, articulation, and repetition. The goal is to create music that grows out of how the instrument behaves and how you interact with it, rather than from narrative ideas or realistic sound references. Think of this piece as an exploration of process, control, and musical behavior, where form emerges through interaction with a sound system you designed.

With this project, you can work with the following material:

1. Pick two sound sources below. For category a, you can pick one type of sound category and record as many different sounds as possible. From category b, you can only pick one object, but you can record different sounds of that object.

For example, if you picked glass and music box, you can record different glass sounds, but you should record only the music box. You can have normal music box sound, you can have only pitchless clicks or just the lever noise etc..

The recordings must be 48K - 24 bit, mono or stereo.

Category a: pick sounds from one category below. You can record any type of sound in the chosen category.

Metal	Glass	Plastic	Ceramic
Singing Bowl	Wine glass	Rubber band	Ceramic Tiles
Metal Spring	Marbles	Balloon	Mugs and cups
Wok	Bottles (w. or w/o water)	Bubble wrap	Flower pots
Coins and Keys	et...	etc...	etc...
etc...			

Category b: pick only one sound below. You can only use one type of sound in the category below.

Mechanical / Time-Based Objects	Plucked / Resonant Mechanical Objects	Typing / Impact / Key-Based Objects	Motorized / Moving Objects	Friction / Surface Interaction Objects	Percussive / Industrial Small Objects
Metronome (acoustic output only)	Music box	Typewriter	Wind-up toys	Paper shredder	Keys (single or bunch)
Clock / Watch (mechanical ticking, winding, alarms)	Jaw harp	Adding machine / calculator with physical buttons	Small battery-powered toys (motors, gears, clicks)	Hole punch	Coins
Kitchen timer	Kalimba / thumb piano (borderline—acceptable if treated as object, not performance)	Cash register	Electric toothbrush	Sandpaper on wood or metal	Screws, bolts, nuts
Old alarm clock	Toy piano (only if treated mechanically, not melodically)	Computer keyboard (acoustic, no MIDI)	Small fan (handheld, desk fan)	Hole punch	Metal washers
Mechanical counter / tally counter	Tuning fork (single or multiple, struck or damped)	Old telephone keypad / rotary dial	CD/DVD drive sounds (tray, spin-up, read errors)	Sandpaper on wood or metal	Small tools (wrenches, pliers, screwdrivers)
Wind-up stopwatch		Mechanical doorbell		Hinges, springs, coils	Office desk accessories (clips, rulers, trays)

- Compose a tape music piece (3–5 minutes) using only the samples you recorded for this project. In addition to working with your sounds as audio files in a fixed-media context, you must also create and use at least one sampler instrument (Kontakt) built from your recordings as part of the composition.
- You may use all the tape effects we used in the last project. In addition to working with your sounds as audio files in a fixed-media context, as a core requirement of this project, you must build a Kontakt instrument from at least one of your sources, and you must use it in your project. This could be either a playable instrument in a traditional sense or a sampling device that lets you control multiple samples at once. Do not forget to use groups and velocity ranges creatively and assign different automations to your sampled instrument.
- Be prepared to present in class on March 23rd.

GRADING

Correct Audio format: 10 pts

No unintended distortion or digital clip: 10 pts

Kontakt instrument: 20 pts

Sound Source rules are followed: 25 pts

Presentation: 25 pts

MSC 487 Introduction to Electronic Music
**PROJECT 4 -ACOUSMATIC COMPOSITION “CINEMA FOR
THE EAR”**

Bilkent University — Department of Music — Spring 2026 — Tolga Yayalar

DEADLINE: 6 April during class

Although narrative or storytelling is one possible approach to this project, it is not required and not expected as the default. You are encouraged to engage with your chosen idea in a variety of ways, including but not limited to abstraction, metaphor, mapping, or atmosphere. Your piece does not need to describe events or follow a literal storyline. Instead, you may translate visual, textual, or conceptual elements into sound through processes such as transformation, association, or sonification (in an artistic, non-data-driven sense). For example, you might interpret colors as spectral regions, movement as changes in density or spatial motion, or written descriptions as sources for sound selection and spatial behavior. You may also focus on creating a particular mood, texture, or sonic environment related to your chosen reference. The emphasis is on how sound is organized, placed in space, and perceived, rather than on representing a story in a literal or illustrative way.

Compose a 3-5-minute-long piece with the following constraints.

1. Record sounds in your environment, which will become elements of your composition. There are no limitations on what you record. Your sound sources may be as varied as you wish. As always, record 48 kHz and 24 bits.
2. Transfer the sounds to the computer and create a sound library by labeling them accordingly.
-edit the sounds and label them accordingly
3. You can use all the studio techniques available to you except, of course, no software instruments or libraries. As always, you must record or create your own sounds to use them in your piece. You cannot use sounds that you didn't record by yourself.
4. Compose a piece of music only using the sounds in your library. You must creatively use **stereo imaging and reverb** to create a sonic space related to your composition.
6. Your composition must be based on one of the ideas shared below. You can pick any one of the ideas and find inspiration. More than one person can pick the same idea. Your inspiration may be in the form of:
 - Atmospheric or perceptual interpretation (mood, density, tension, stillness, immersion)
 - Symbolic or metaphorical translation
 - Structural or formal mapping (e.g., form, proportion, transformation, repetition)
 - Cross-modal interpretation based on color, movement, shape, or texture
 - Narrative or storytelling approaches, understood in a non-literal or suggestive sense

7. Present your piece in class on April 6th
8. Write down a one-paragraph (250 words) program note. Explain how your piece fits into the concept you are using.

IDEAS

VIDEO ART AND OTHER INSTALLATIONS

1. Bill Viola - Reflecting Pool (1977) - <https://www.youtube.com/watch?v=Mb1hL5P2ztU>
2. Andy Goldsworthy - Rivers and Tides (watch between 11:00 - 18:00) - <https://www.dailymotion.com/video/x15e83f>

PAINTINGS AND PRINTS

3. M.C. Escher, "Metamorphosis II" https://www.reddit.com/r/printmaking/comments/3w5mml/mc_escher_metamorphosis_ii_1940/#lightbox
4. René Magritte, The Human Condition (1933) - <https://www.renemagritte.org/the-human-condition.jsp>
5. Enrico Bartzago, Colourful Village Life https://upload.wikimedia.org/wikipedia/commons/3/32/Enrico_Bartzago_-_Colourful_Village_Life.jpg

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6. Emad Nassar, "Gaza Bath Time" <https://time.com/wissam-nassar-at-home-in-gaza/>
7. Andrea Gursky, "99-Cent-II-Diptychon" <https://i0.wp.com/blog.artsper.com/wp-content/uploads/2013/07/New-Featured-Image-1200-x-675-3.jpg?fit=1200%2C675&ssl=1>

LITERATURE

8. Orhan Pamuk, Kara Kitap-Boğazın Suları Çekildiği Zaman - <https://sessiziz.wordpress.com/2010/03/27/bogazin-sulari-cekildigi-zaman/>
9. Robert Frost, "Stopping by Woods on a Snowy Evening" <https://www.poetryfoundation.org/poems/42891/stopping-by-woods-on-a-snowy-evening>
10. Italo Calvino, "Invisible Cities Cities and the Dead: Eusapia" page 109-110 https://www.dropbox.com/s/v3m5b3l0tid1109/Invisible_cities-Calvino.pdf?dl=0
11. T.S. Elliott, The Waste Land: I. The Burial of the Dead <https://www.poetryfoundation.org/poems/47311/the-waste-land>

CONCEPTS

12. The concept of Pareidolia. <https://www.artsandmindlab.org/pareidolia-research/>
13. The concept of Entropy as a fundamental concept that dictates the one-way direction of time. According to the Second Law of Thermodynamics, entropy—or the measure of disorder in a system—always tends to increase in an isolated system. This means that natural processes move from a state of order to a state of disorder over time. <https://sciencenotes.org/what-is-entropy-definition-and-examples/>
14. Figure-ground perception is a principle of perception describing how sensory information is organized by distinguishing a focal element (the figure) from its surrounding context (the ground) <https://www.brainbalancecenters.com/blog/visual-processing-explained-visual-figure-ground-perception#:~:text=Figure%2Dground%20perception%20is%20the,rather%20than%20the%20paper's%20texture>

GRADING

Correct Audio format: 10 pts

No unintended distortion or digital clip: 10 pts

Conceptual idea: 20 pts

Sound Source rules are followed: 25 pts

Presentation: 25 pts

MSC 487 Introduction to Electronic Music
PROJECT V -MULTI-CHANNEL DIFFUSION

Bilkent University — Department of Music — Spring 2026 — Tolga Yayalar

DEADLINE: 27 April during class

Ambisonic Spatial Reinterpretation (Multi-Channel Diffusion)

This project focuses on multi-channel spatial diffusion through the creation of a high-order ambisonic rendering of an existing stereo electroacoustic work. Students will select a stereo piece from the course repertoire and reinterpret its spatial dimension by deploying the original stereo image into a three-dimensional ambisonic space. Rather than live diffusion, this project emphasizes composed spatial trajectories realized through automation.

Before working in ambisonics, students must devise a clear spatial plan, outlining how space functions structurally throughout the piece and how spatial parameters contribute to form, gesture, and perception. Using automation, students will articulate movement, depth, and spatial transformation while preserving the internal coherence and integrity of the original stereo image. The completed ambisonic rendering will be adapted to the department's playback system and presented in class.

Rules:

- No effects may be used other than the spatial tools introduced in class.
- The original stereo mix must be preserved and respected; its internal spatial relationships should remain intact, even as the stereo image is expanded and deployed into three-dimensional space.
- Spatialization must function as an **integral structural element** of the piece. Spatial decisions should be intentional and well planned—avoid random movement, unstructured gestures, or spatial overcrowding.

GRADING

Correct Audio format: 10 pts

No unintended distortion or digital clip: 10 pts

Sound Source rules are followed: 55 pts

Presentation: 25 pts